Windows Logs Monitoring & Analysis

Using Windows Event Viewer

Cybersecurity Project Report

Subject: Cybersecurity

Tool: Windows Event Viewer

Platform: Microsoft Windows | Computer: Abbas

By: Muhammad Afaq Abbas

1. Introduction

Everything we perform on Windows is being recorded. every login attempt, every service change, every hardware event, every application that writes something worth noting. There is no setup is required for this. It happens by itself in the background.

This project involves looking at the window logs. Using the Windows event viewer on the machine named Abbas. The three main types of Windows logs were examined one by one: security, system, and Application. Events Viewer was opened and took screenshots of different events as evidence that had happened, the objective of which was practical understanding. And actual familiarity with what logs contain and what that information is good for.

2. Objectives

This project worked for five specific objectives.

- Understanding what the Windows logs are and why they are automatically generated.
- Learn to navigate Event Viewer and locate the three main log categories.
- Read logs and analyze log entries with the support of screenshots.
- Understand what the Event IDs encountered during analysis are
- Recognize the connection between log monitoring and real security and administration work.

3. Tool — Windows Event Viewer

No installation is required for the Windows Event Viewer. It is the built-in diagnostic tool that can be opened by typing Win + R and then “eventvwr.msc” in Run. The left panel lists all log categories. Three entries appear under the Windows logs: Security, System, and Application. All events are written automatically by the operating system over time. Each held tens of thousands of events.

Each event follows a consistent format: an Event ID labeling what happened, a Source identifying what generated it, a Level indicating severity, an exact timestamp, and a plain-text description. Five fields. Usually enough to understand the full picture of any given event.

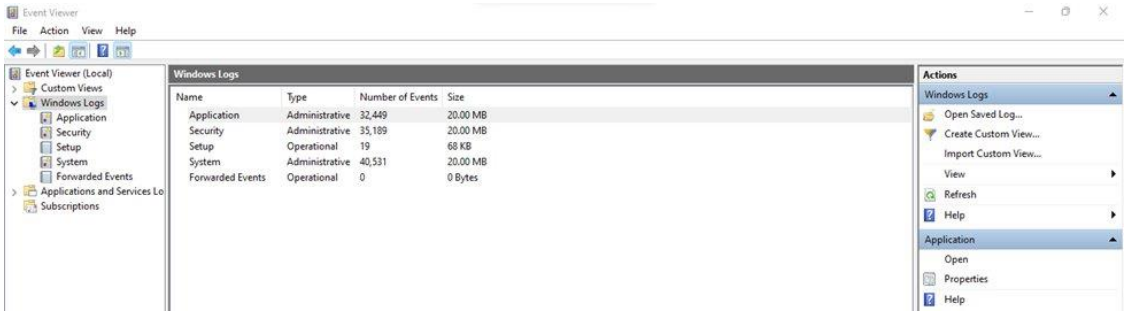


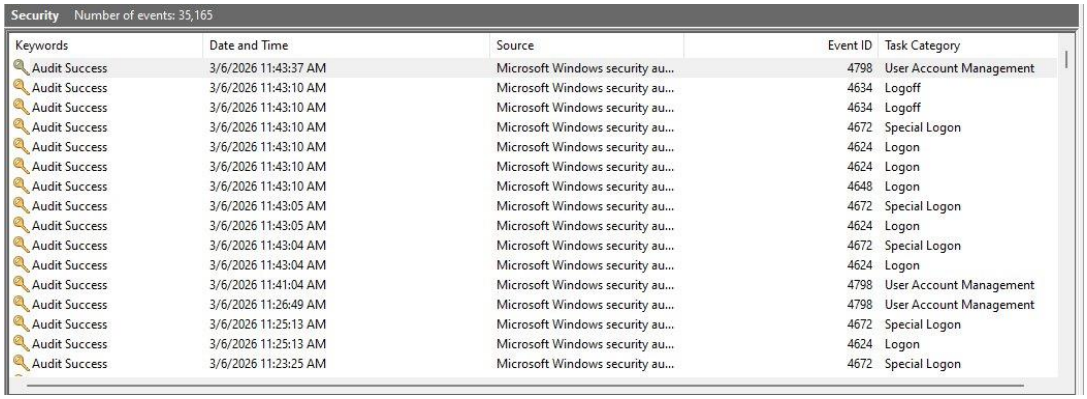
Figure 1: Event Viewer — Windows Logs Overview (Security: 35,189 | System: 40,531 | Application: 32,449 events)

4. Security Log

Every login attempt, logoff, privilege assignment, and account change on the system is being tracked by the security log authentication. Each entry is marked either Audit Success or Audit Failure. Security analysts treat this log as the first place to check when something suspicious may have happened. This project, it contained 35,165 events. Nearly all were Audit Success. One was not, and it deserves attention.

Event ID	Category	What It Means	Why It Matters
4624	Logon	Account successfully logged on	Tracks every successful login
4625	Logon	Account failed to log on	Detects wrong passwords / brute force
4634	Logoff	Account logged off	Confirms session ended normally
4672	Special Logon	Admin privileges assigned at logon	Tracks elevated access
4798	Account Mgmt	User group membership enumerated	Detects reconnaissance activity

4.1 Security Log — List View



Security Number of events: 35,165

Keywords	Date and Time	Source	Event ID	Task Category
Audit Success	3/6/2026 11:43:37 AM	Microsoft Windows security au...	4798	User Account Management
Audit Success	3/6/2026 11:43:10 AM	Microsoft Windows security au...	4634	Logoff
Audit Success	3/6/2026 11:43:10 AM	Microsoft Windows security au...	4634	Logoff
Audit Success	3/6/2026 11:43:10 AM	Microsoft Windows security au...	4672	Special Logon
Audit Success	3/6/2026 11:43:10 AM	Microsoft Windows security au...	4624	Logon
Audit Success	3/6/2026 11:43:10 AM	Microsoft Windows security au...	4624	Logon
Audit Success	3/6/2026 11:43:10 AM	Microsoft Windows security au...	4648	Logon
Audit Success	3/6/2026 11:43:05 AM	Microsoft Windows security au...	4672	Special Logon
Audit Success	3/6/2026 11:43:05 AM	Microsoft Windows security au...	4624	Logon
Audit Success	3/6/2026 11:43:04 AM	Microsoft Windows security au...	4672	Special Logon
Audit Success	3/6/2026 11:43:04 AM	Microsoft Windows security au...	4624	Logon
Audit Success	3/6/2026 11:41:04 AM	Microsoft Windows security au...	4798	User Account Management
Audit Success	3/6/2026 11:26:49 AM	Microsoft Windows security au...	4798	User Account Management
Audit Success	3/6/2026 11:25:13 AM	Microsoft Windows security au...	4672	Special Logon
Audit Success	3/6/2026 11:25:13 AM	Microsoft Windows security au...	4624	Logon
Audit Success	3/6/2026 11:23:25 AM	Microsoft Windows security au...	4672	Special Logon

Figure 2: Security Log — 35,165 events, all Audit Success (Event IDs: 4624, 4634, 4672, 4798)

4.2 Event ID 4624 — Successful Logon

This event recorded account ABBASS from the WORKGROUP domain logging in at 11:43:10 AM on 3/6/2026. Logon Type 2 means it was a direct interactive session at the machine itself. Result: Audit Success on computer Abbas. Routine, expected, and informative — it establishes exactly who was on the system and when the session began.

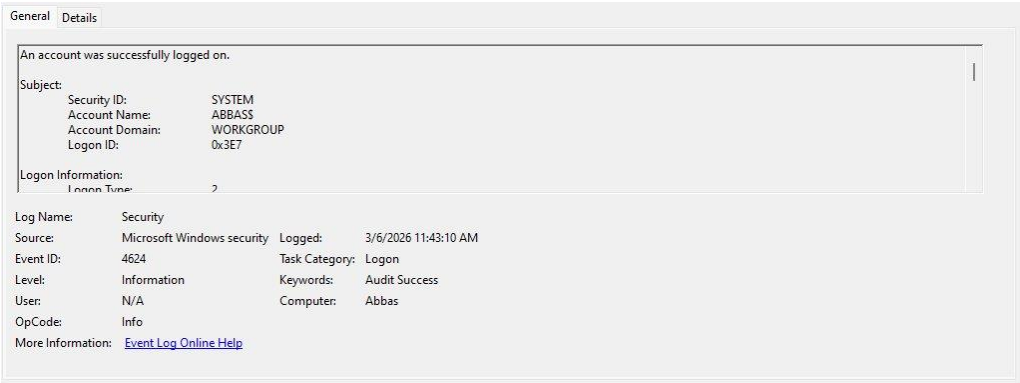


Figure 3: Event ID 4624 — Successful Logon Detail (Account: ABBASS)

4.3 Event ID 4634 — Account Logoff

At the same timestamp — 11:43:10 AM — account ABBAS\Dell logged off. The matching Logon ID in both events ties them together as a single session. This pairing of 4624 and 4634 is how Windows documents the full lifecycle of any login: when it started, and when it ended. A clean session with no gaps.

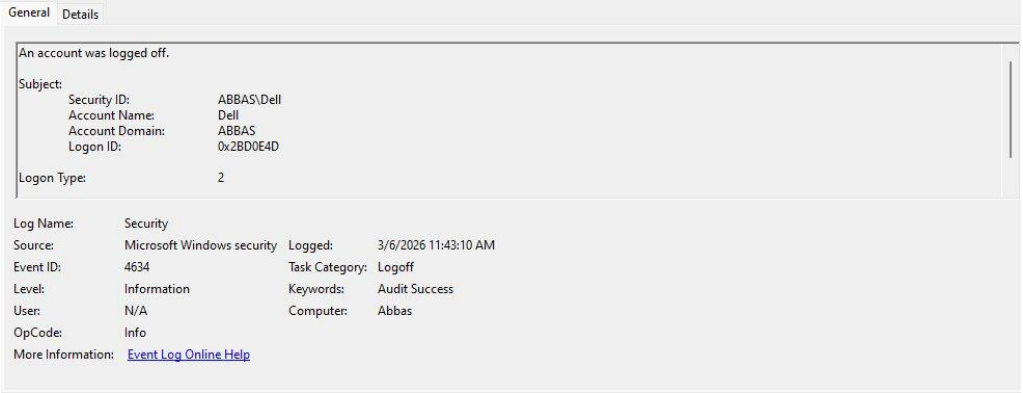


Figure 4: Event ID 4634 — Account Logoff Detail (Account: Dell, Domain: ABBAS)

4.4 Event ID 4625 — Failed Logon (Audit Failure)

Recorded at 11:01:50 AM on 3/6/2026, this was the one Audit Failure in the entire Security log. The cause: a wrong password entered during a login attempt. On its own, one failure means nothing — it happens. What makes Event ID 4625 significant is the pattern and volume. A burst of these events within seconds or minutes is a reliable indicator of a brute force attack. One is forgettable. Fifty in two minutes is not.

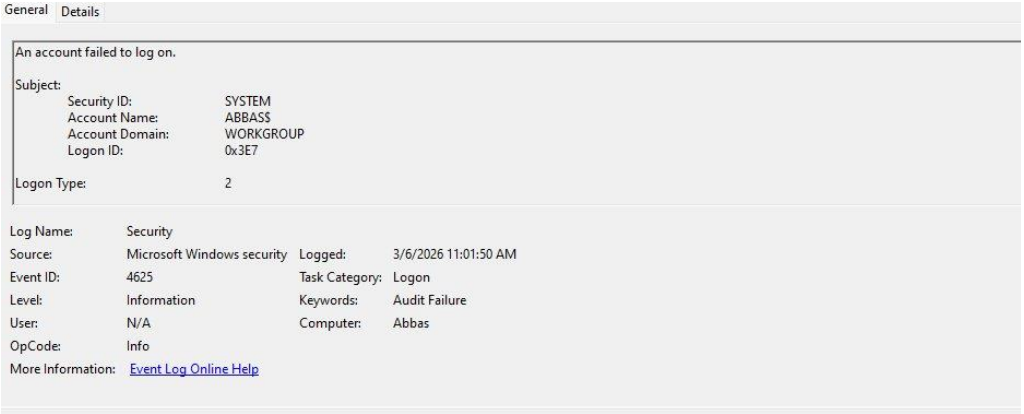


Figure 5: Event ID 4625 — Failed Logon Detail (Audit Failure — the only failure in the log)

5. System Log

The System log records operating system activity — kernel events, hardware changes, driver behavior, service states. With 40,469 entries, it was the largest of the three logs examined. The majority were informational, documenting ordinary system behavior. A handful of Warnings appeared. And one Error stood out as the most actionable discovery in the entire project.

Event ID	Source	What It Means	Why It Matters
6013	EventLog	System uptime reported (seconds)	Shows system stability
105	Kernel-Power	Power source changed	Tracks AC/battery transitions
1	Kernel-General	Kernel operational / boot event	Confirms successful OS start
566	Kernel-Power	Hardware power state change	Monitors hardware events
2505	Server	Network transport binding failed	Network naming conflict (Error)

5.1 System Log — List View

System Number of events: 40,469 (!) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	3/6/2026 12:00:00 PM	EventLog	6013	None
Information	3/6/2026 11:53:08 AM	Kernel-Power	105	(100)
Error	3/6/2026 11:43:08 AM	Server	2505	None
Information	3/6/2026 11:43:05 AM	Power-Troubleshooter	1	None
Information	3/6/2026 11:43:05 AM	Kernel-Power	566	(268)
Information	3/6/2026 11:43:05 AM	BTHUSB	18	None
Information	3/6/2026 11:43:04 AM	Kernel-Power	566	(268)
Information	3/6/2026 11:43:04 AM	Kernel-Power	131	(33)
Information	3/6/2026 11:43:04 AM	Kernel-General	1	(5)
Information	3/6/2026 11:41:05 AM	Kernel-Power	107	(102)
Information	3/6/2026 11:41:04 AM	Kernel-Power	42	(64)
Information	3/6/2026 11:41:04 AM	Kernel-Power	566	(268)
Warning	3/6/2026 11:28:06 AM	DNS Client Events (Microsoft...	1014	(1014)
Warning	3/6/2026 11:25:05 AM	DNS Client Events (Microsoft...	1014	(1014)
Information	3/6/2026 11:23:35 AM	Kernel-General	15	(10)
Information	3/6/2026 11:23:32 AM	Kernel-General	16	None
Information	3/6/2026 11:23:32 AM	Kernel-General	16	None
Information	3/6/2026 11:23:32 AM	Kernel-General	16	None
Information	3/6/2026 11:23:31 AM	Kernel-General	16	None

Figure 6: System Log — 40,469 events (Info, Warning, and Error levels visible)

5.2 Event ID 6013 — System Uptime

The EventLog service writes this entry once every 24 hours. The recorded value was 170,584 seconds — approximately 47.4 hours, or just under two full days of continuous operation. No reboots, no crashes. For a monitoring scenario, stable uptime over this period reflects a healthy, uninterrupted machine. An unexpected drop in this number is always worth investigating.

General Details

The system uptime is 170584 seconds.

Log Name:	System		
Source:	EventLog	Logged:	3/6/2026 12:00:00 PM
Event ID:	6013	Task Category:	None
Level:	Information	Keywords:	Classic
User:	N/A	Computer:	Abbas
OpCode:	Info		
More Information:	Event Log Online Help		

Figure 7: Event ID 6013 — System Uptime: 170,584 seconds (Source: EventLog)

5.3 Event ID 105 — Power Source Change

At 11:53:08 AM, the Kernel-Power component logged a transition between AC power and battery. The SYSTEM account generated this automatically, with no user action involved. On a laptop, power source changes are routine. Their value is diagnostic — when investigating an unexpected shutdown or battery fault, an Event 105 entry provides an exact timestamp for when the power state changed.

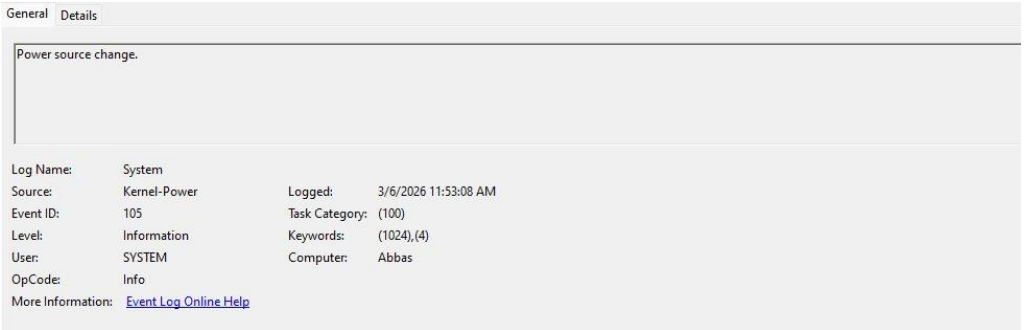


Figure 8: Event ID 105 — Power Source Change (Source: Kernel-Power, User: SYSTEM)

5.4 Event ID 2505 — Network Error ⚠

This was the most significant finding in the project. The Server service reported that it could not bind to the network transport because another machine on the same network was already using the hostname Abbas. This is a NetBIOS naming conflict — two devices sharing an identical name create a collision that prevents the server service from initializing. The fix is straightforward: rename one device. But without Event 2505 pointing directly at the problem, this failure would have been very difficult to trace. The log made an invisible issue visible.

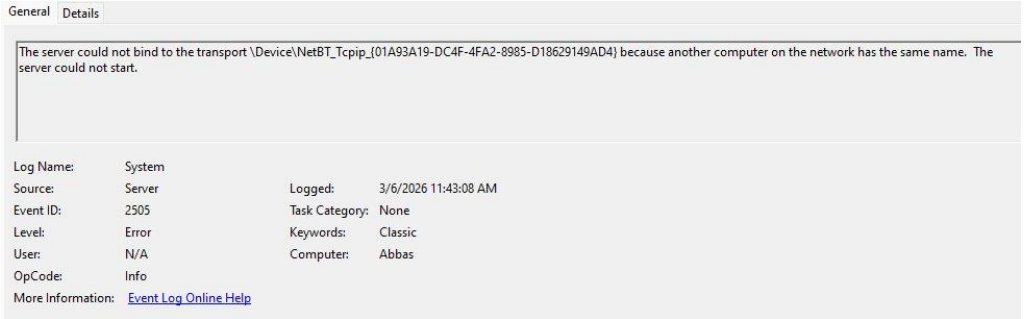


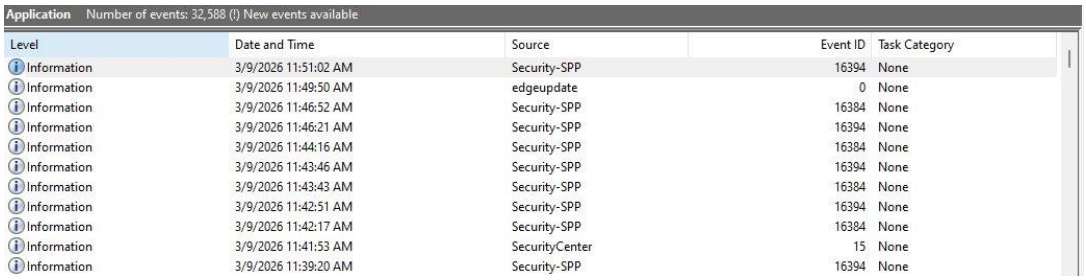
Figure 9: Event ID 2505 — Network Transport Binding Error (Error Level — Source: Server)

6. Application Log

The Application log collects events from software running on the system. The dominant source here was Security-SPP, the Software Protection Platform, which manages Windows licensing in the background. Every event in the observation window was at the Information level. No application errors, no crashes — just the system's software layer running as expected.

Event ID	Source	What It Means	Why It Matters
16384	Security-SPP	SPP service restart scheduled	Windows licensing refresh
16394	Security-SPP	Licensing migration succeeded	Policy applied successfully
0	edgeupdate	Generic unregistered app event	Source context needed to interpret
15	SecurityCenter	Security Center status notification	Confirms AV/firewall active

6.1 Application Log — List View



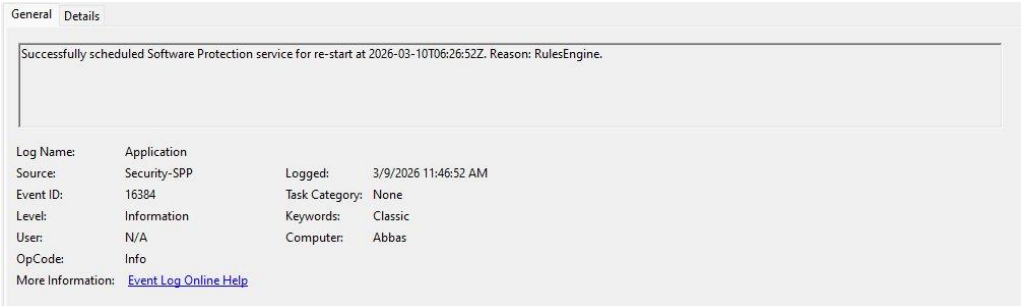
Application Number of events: 32,588 (!) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	3/9/2026 11:51:02 AM	Security-SPP	16394	None
Information	3/9/2026 11:49:50 AM	edgeupdate	0	None
Information	3/9/2026 11:46:52 AM	Security-SPP	16384	None
Information	3/9/2026 11:46:21 AM	Security-SPP	16394	None
Information	3/9/2026 11:44:16 AM	Security-SPP	16384	None
Information	3/9/2026 11:43:46 AM	Security-SPP	16394	None
Information	3/9/2026 11:43:43 AM	Security-SPP	16384	None
Information	3/9/2026 11:42:51 AM	Security-SPP	16394	None
Information	3/9/2026 11:42:17 AM	Security-SPP	16384	None
Information	3/9/2026 11:41:53 AM	SecurityCenter	15	None
Information	3/9/2026 11:39:20 AM	Security-SPP	16394	None

Figure 10: Application Log — 32,588 events (Security-SPP, edgeupdate, SecurityCenter)

6.2 Event ID 16384 — SPP Service Restart Scheduled

The Software Protection Platform used this event to document a planned future action: a service restart scheduled for 2026-03-10T06:26:52Z, triggered by the RulesEngine. Windows recorded the intent before executing it. This pattern — planning ahead and logging the plan — is characteristic of how Windows manages its own internal maintenance without any user involvement.



General Details

Successfully scheduled Software Protection service for re-start at 2026-03-10T06:26:52Z. Reason: RulesEngine.

Log Name:	Application		
Source:	Security-SPP	Logged:	3/9/2026 11:46:52 AM
Event ID:	16384	Task Category:	None
Level:	Information	Keywords:	Classic
User:	N/A	Computer:	Abbas
OpCode:	Info		
More Information:	Event Log Online Help		

Figure 11: Event ID 16384 — SPP Service Restart Scheduled (Source: Security-SPP)

6.3 Event ID 16394 — Licensing Migration Succeeded

Where 16384 documented the plan, 16394 documented the outcome: offline downlevel migration succeeded. Together, these two events represent one complete licensing compliance cycle — intent recorded, action completed, result confirmed. Seeing them appear in sequence is evidence that the Software Protection Platform is functioning correctly.

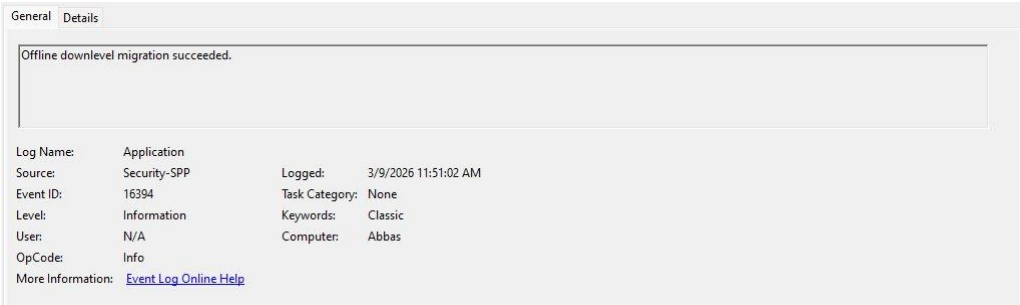


Figure 12: Event ID 16394 — Offline Downlevel Migration Succeeded (Source: Security-SPP)

6.4 Event ID 0 — Generic edgeupdate Event

Event ID 0 is what appears when an application writes to the log without registering its event descriptions with Windows first. The description field says only that no matching description was found. This is not an error — it is a registration gap. The Microsoft Edge update service generated this entry. The source name, edgeupdate, is the only useful identifier, and in this case, it is enough.

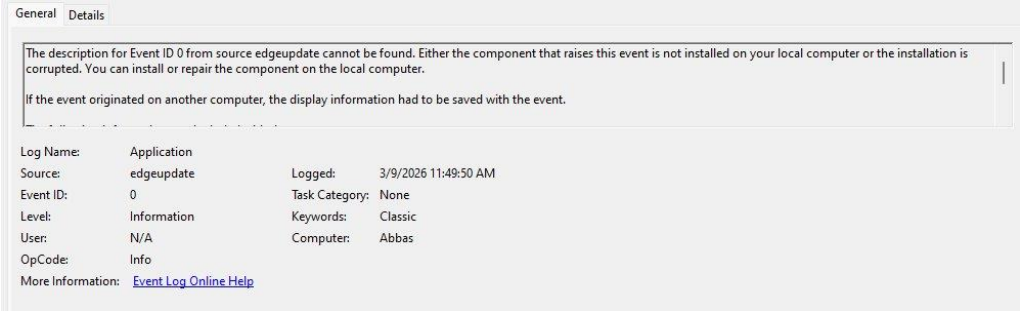


Figure 13: Event ID 0 — Generic Event: Description not registered (Source: edgeupdate)

7. Key Findings & Observations

The three logs each produced distinct and useful findings:

Security Log

- 35,164 Audit Success events to 1 Audit Failure — the authentication record is clean, with one failed login noted at 11:01:50 AM
- Events 4624 and 4634 shared an identical timestamp, confirming session tracking is accurate, and the session lifecycle was properly recorded
- A single 4625 event is unremarkable on its own; the significance of failed logins lies in volume and frequency, not in isolated occurrences

System Log

- An uptime of 47.4 hours confirmed stable, uninterrupted machine operation across the observation period
- Event 2505 was the project's most actionable discovery: a real NetBIOS naming conflict actively blocking a service from starting
- Power and kernel events were consistent with normal laptop operation and required no corrective action

Application Log

- The 16384 and 16394 event pair from Security-SPP confirmed a complete and successful licensing cycle
- Event ID 0 from edgeupdate illustrated what happens when applications skip event registration — the source name becomes the only available context
- All entries were at the Information level, indicating no application faults during the observation window

8. Advantages of Log Monitoring

This project illustrated in concrete terms why log monitoring matters. The reasons are practical:

- **Security:** Threats leave a trace immediately — failed logins, unauthorized attempts, and privilege misuse all appear in the Security log as they happen
- **Troubleshooting:** Silent failures become diagnosable — the naming conflict behind Event 2505 was invisible at the surface level but clearly documented in the log
- **Forensics:** Authentication activity is permanently timestamped and attributed — who logged in, when, from where, and under what account
- **Compliance:** Compliance frameworks, including PCI-DSS, HIPAA, and ISO 27001, mandate security event logging — these logs satisfy that requirement directly

- **Monitoring:** Long-term system behavior becomes trackable — uptime trends, recurring faults, service state changes, and hardware events build a reliable picture over time

9. Conclusion

What this project demonstrated is simple but important: Windows is already collecting detailed, accurate information about everything happening on a machine. It does not need to be asked. The only variable is whether that information gets read. This project read it, and the results were not theoretical. Event 2505 exposed a real network conflict. Event 4625 documented a real failed login. Event 6013 confirmed real uninterrupted uptime. These findings came from an ordinary machine running its ordinary day.

Each log answered a different category of question. The Security log addressed authentication: who logged in, whether anything failed, and whether anything was unusual. The System log addresses infrastructure: is the OS and hardware behaving correctly, and are services running? The Application log addresses software: what is running in the background, and is it completing its tasks? Together, the three logs give a reasonably complete operational picture of any Windows system.

Reading logs is not a specialist skill. It is a foundational one. Every IT and cybersecurity role encounters logs in some form. Knowing how to open Event Viewer, interpret Event IDs, and distinguish normal behavior from something worth investigating is a baseline competency — and this project is a reasonable starting point for building it. The logical next steps involve learning to filter and search logs efficiently, creating custom views for specific Event IDs, and understanding how enterprise environments centralize and automate log analysis through SIEM platforms.

10. References

Microsoft Corporation. (2024). Windows Event Viewer overview. https://docs.microsoft.com/en-us/windows/security/threat-protection/auditing/

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Stallings, W. (2019). Operating Systems: Internals and Design Principles (9th ed.). Pearson.

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